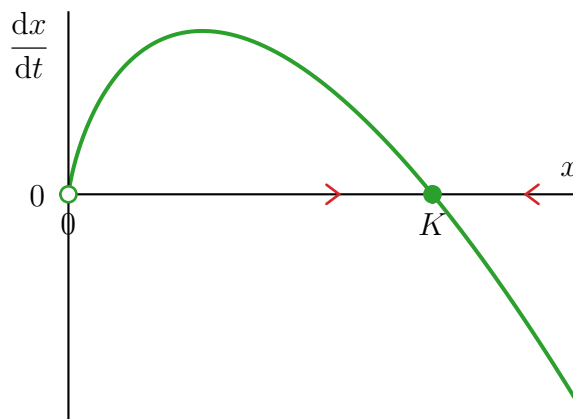


PROBLEM SHEET 1: SOLUTIONS

1. (a) The phase diagram looks like



and so we have equilibria:

- $x = 0$, unstable,
- $x = K$, stable.

(You might argue that you can't evaluate dx/dt at $x = 0$ because of the $\log(K/x)$ term. But as the graph illustrates, $\lim_{x \rightarrow 0} \log(K/x) = 0$, and for the purposes of biological population modelling, this is enough.)

- (b) This is very similar to the logistic model. It's Kolmogorov, so populations can't grow out of nothing (from $x = 0$). We once again have a carrying capacity $x = K$ and the parameter a again really just scales time. The difference is that the phase space is no longer symmetric: the Gompertz model shuts down growth more rapidly as x approaches the carrying capacity.
2. (a) This is a logistic model – with carrying capacity K and time scaling R – but with an additional constant death rate P . The P makes this a simple *harvesting* model: imagine something like a fishing quota, where a fixed number of fish are removed every week. Obviously this model requires a sufficient number of the population to exist, though, in order for x not to become negative. We will see this model again later in the course.
- (b) At $x = 0$, dx/dt is negative and this is nonsense.
- (c) The key here is to see that the left-hand side has units density/time and so the right-hand side needs to match. Furthermore, you can't add two terms with different units, so anything added together needs to match units. For this to work:
- R needs units of 1/time,
 - K needs units of density,
 - P needs units of density/time,
 - A needs units of density.

(d) Letting $x = X\hat{x}$ and $t = T\hat{t}$, substituting this into our model we have

$$\frac{X}{T} \frac{d\hat{x}}{d\hat{t}} = RX\hat{x} \left(1 - \frac{X\hat{x}}{K}\right) - P \left(1 - \exp \left[-\frac{X^2\hat{x}^2}{\alpha A^2}\right]\right). \quad (1)$$

Fixing the left-hand side first, we multiply through to get

$$\frac{d\hat{x}}{d\hat{t}} = RT\hat{x} \left(1 - \frac{X\hat{x}}{K}\right) - \underbrace{\frac{PT}{X}}_{[2]} \underbrace{\left(1 - \exp \left[-\frac{X^2\hat{x}^2}{\alpha A^2}\right]\right)}_{[1]}. \quad (2)$$

Keeping our eye on the prize, we tackle term [1] first to set $X = A$.

Then we look at term [2], and set $T = A/P$. This gives us

$$\frac{d\hat{x}}{d\hat{t}} = \frac{RA}{P} \hat{x} \left(1 - \frac{A}{K} \hat{x}\right) - \left(1 - \exp \left[-\frac{\hat{x}^2}{\alpha}\right]\right), \quad (3)$$

and so we can directly say $r = RA/P$ and $q = K/A$, giving us the nondimensionalised equation.

In conclusion,

$$X = A, \quad T = \frac{A}{P}, \quad r = \frac{RA}{P}, \quad q = \frac{K}{A}. \quad (4)$$

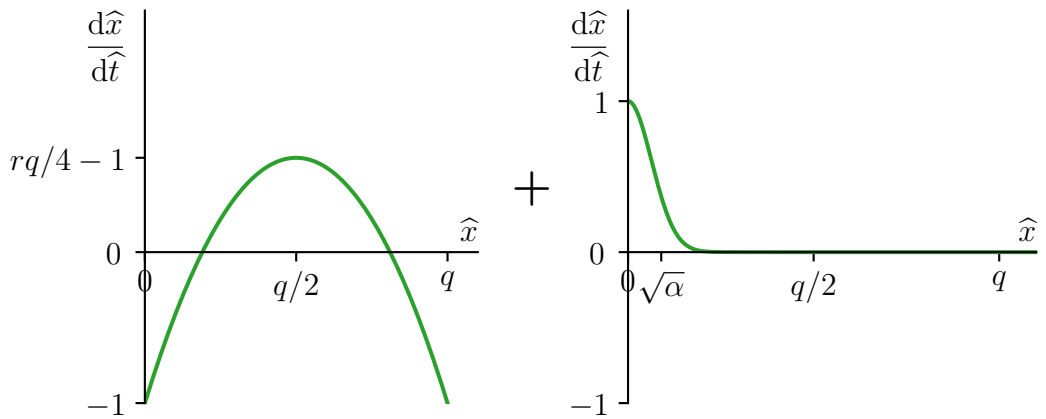
(e) The equation we're trying to sketch is

$$\frac{d\hat{x}}{d\hat{t}} = \underbrace{r\hat{x} \left(1 - \frac{\hat{x}}{q}\right)}_{(i)} - \underbrace{\left(1 - \exp \left[-\frac{\hat{x}^2}{\alpha}\right]\right)}_{(ii)}, \quad (5)$$

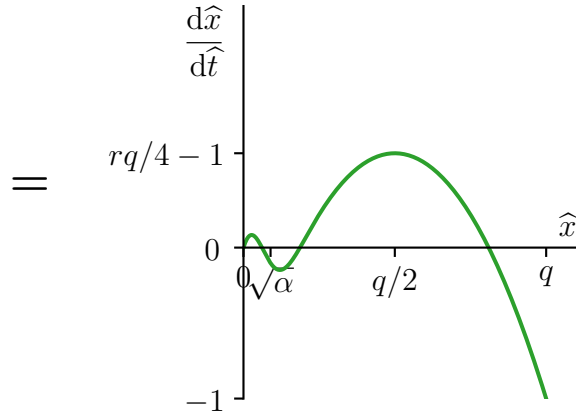
This is the sum of two bits:

- (i) A logistic ‘unhappy quadratic’, shifted down by 1. This was a problem because $d\hat{x}/d\hat{t} = -1 < 0$ at $\hat{x} = 0$. (See below left.)
- (ii) The exponential term, which you might recognise as a Gaussian with standard deviation $\sqrt{\alpha}$. The standard deviation is proportional to what we might call the ‘width’ of the Gaussian. (see below right.) Notice how, since $\alpha \ll 1$, it acts far away from the peak of the logistic.

Adding the two graphs together, then, we get



which



So we can see the problem is fixed, since now $\hat{x} = 0$ is also an equilibrium.

- (f) Looking at the phase space, it's clear there are at least two nonzero equilibria if the top of the unhappy quadratic is positive.

But you can also see there is another nonzero equilibrium near to $\sqrt{\alpha}$. You might not be able to see this in your own sketch, but mathematically we can see it has to be there because the slope of the phase space plot,

$$\frac{d}{d\hat{x}} \left[\frac{d\hat{x}}{d\hat{t}} \right] = r \left(1 - \frac{2\hat{x}}{q} \right) + \frac{2\hat{x}}{\alpha} \exp \left[-\frac{\hat{x}^2}{\alpha} \right], \quad (6)$$

evaluated at $\hat{x} = 0$, equals $r > 0$. So the slope at the origin is positive and since the function is continuous, we expect another equilibrium as the graph goes beneath the $\hat{x}$ -axis for increasing $\hat{x}$.

Getting all three equilibria therefore requires that the top of the unhappy quadratic is positive. Since $\alpha \ll 1$, the effect of the exponential is gone by the peak of the quadratic, so we can just say

$$0 = r \left(1 - \frac{2\hat{x}}{q} \right) \implies \hat{x} = q/2. \quad (7)$$

At this value of $\hat{x}$, we have $d\hat{x}/d\hat{t} = rq/4 - 1$, as seen in the graph of the full model above. We need this to be positive, and so

$$\frac{rq}{4} - 1 > 0 \implies rq > 4. \quad (8)$$

3. (a) The first term of (7) is a growth term. Alone it would be

$$\frac{dx}{dt} = x \quad (9)$$

which gives exponential growth. The same is true for the term $2y$ in (8). This is quite standard.

The second term in (7), $-2y$ will cause the population x to decrease, presumably as the predator y kills the prey. But this term has no dependence on the size of the population, x , unlike in Lotka–Volterra. This seems unrealistic as it should be harder for the predator to act if the prey size x is small (and vice versa if large). Similar analysis holds for the $2x$ term in (8).

Finally, the constant terms 4 and -10 imply population-independent growth, maybe some kind of source or sink, like immigration or emigration.

Overall this is a bad model for a predator–prey system.

(b) We need to solve the linear system

$$x - 2y + 4 = 0 \tag{10}$$

$$2x + 2y - 10 = 0. \tag{11}$$

Adding the equations we get

$$3x + -6 = 0 \implies x = 2 \tag{12}$$

which can then be used to get $y = 3$. This is permissible.

The Jacobian of the system is

$$\mathbf{J} = \begin{pmatrix} 1 & -2 \\ 2 & 2 \end{pmatrix}, \tag{13}$$

so we need to find its eigenvalues. We do so by solving

$$\det(\mathbf{J} - \lambda \mathbf{I}) = 0 \implies (1 - \lambda)(2 - \lambda) + 4 = 0 \tag{14}$$

$$\implies \lambda^2 - 3\lambda + 6 = 0, \tag{15}$$

giving us

$$\lambda = \frac{1}{2} \left(3 \pm i\sqrt{15} \right). \tag{16}$$

So both solutions have positive real part and the system is an **unstable spiral**.